

Midterm 2

● Graded

Student

Yahir Cordova

Total Points

87 / 100 pts

Question 1

Question 1

24 / 30 pts

1.1 a

4 / 10 pts

✓ - 0 pts Correct

✓ - 2 pts first one is wrong

✓ - 2 pts second is wrong

✓ - 2 pts third is wrong

- 2 pts fourth is wrong

- 2 pts fifth is wrong

1.2 b

10 / 10 pts

✓ - 0 pts Correct

- 3 pts part a missing

- 2 pts part b missing or wrong

- 3 pts part c missing

- 2 pts part d missing

- 2 pts calculation error

1.3 c

10 / 10 pts

✓ - 0 pts Correct

- 10 pts wrong or not done

- 5 pts conduction is missing

- 5 pts conduction equation is wrong

Question 2

Question 2

30 / 35 pts

2.1 a,b,c

6 / 6 pts

✓ - 0 pts Correct

- 1 pt $z=1$ because it is ideal gas
- 1 pt closed system is missing
- 1 pt energy balance is wrong
- 6 pts not done

2.2 d

4 / 4 pts

✓ - 0 pts Correct

- 1 pt $v_1 = 0.143 \text{ m}^3/\text{kg}$
- 1 pt $v_2 = 0.858 \text{ m}^3/\text{kg}$
- 1 pt should be not linear
- 1 pt $P_2 = 87.35 \text{ kPa} = 0.87 \text{ bar}$
- 1 pt unit error

2.3 e

10 / 10 pts

✓ - 0 pts Correct

- 8 pts work equation is wrong
- 2 pts calculation error
- 2 pts P_2 is wrong
- 10 pts not done

2.4 f

10 / 15 pts

- 0 pts Correct

✓ - 5 pts part a is wrong

- 5 pts part b is wrong
- 5 pts part c is wrong
- 15 pts not done or wrong
- 3 pts ΔT is wrong, it is not zero
- 2 pts C_v for part c is wrong . R should be $8.314/28.9$
- 1 pt calculation error

Question 3

Question 3

33 / 35 pts

3.1 — a,b,c

7 / 7 pts

✓ - 0 pts Correct

- 1 pt control volume
- 3 pts Energy balance is wrong
- 1 pt energy balance work cant be cancelled

3.2 — d

8 / 8 pts

✓ - 0 pts Correct

- 2 pts $v=0.651\text{m}^3/\text{kg}$
- 1 pt calculation error
- 8 pts not done or wrong
- 2 pts not ideal gas

3.3 — e

10 / 10 pts

✓ - 0 pts Correct

- 10 pts not done or wrong
- 1 pt $h_1=3022$
- 1 pt $h_2=2782$
- 1 pt Q should be negative
- 1 pt Q unit conversion
- 3 pts $\dot{m} \text{ times } (h_1-h_2)$
- 1 pt unit conversion
- 1 pt calculation error

3.4 — f

3 / 5 pts

- 0 pts Correct

✓ - 1 pt $v_2=0.651$

✓ - 1 pt $v_1=0.1254$

- 5 pts not done or wrong
- 2 pts v are not marked

3.5  g

5 / 5 pts

✓ - 0 pts Correct

- 5 pts not done

NAME: Vahir CordovaPlease fill in answers in the space supplied and show all your work to receive full credit.

1. a) (10 pts) State whether the following statements are True or False

T If a system undergoes a polytropic process isothermally, $n=1$ for all cases.F Internal energy is a state property.F Heat transfer by convection has the unit of joules. *F A nozzle is a flow passage of varying cross-sectional area in which the velocity and the pressure of a gas or liquid increases in the direction of flow.T Enthalpy for the inlet of the turbine is greater than the outlet of the turbine.b) (10 pts) Determine the volume, in m^3 occupied by 7kg of water at 20MPa and 420°C, using

a. data from compressibility chart

b. data from steam table

c. assuming ideal gas

d. calculate the percent error

b.) @ 200 bar, 420°C

Table A-4

$$v = 0.01108 \text{ m}^3/\text{kg}$$

$$\left(\frac{0.01222 - 0.00994}{440 - 400} \right) \cdot 20 + 0.00994 = 0.01108 \text{ m}^3/\text{kg}$$

$$7 \text{ kg} = 0.0776 \text{ m}^3$$

$$c.) V = nRT \rightarrow V = 0.1119 \text{ m}^3$$

$$d.) PV = znRT \rightarrow 200 \text{ bar} \cdot V = z \cdot 18.02 \text{ kg} \cdot R \cdot 693 \text{ K}$$

$$\Rightarrow V = \frac{znRT}{P} = \frac{0.388 \cdot 18.02 \text{ kg} \cdot 8.314 \text{ J/mol}\cdot\text{K} \cdot 693 \text{ K}}{20000 \text{ kPa}} = 0.1119 \text{ m}^3$$

$$V = 0.1119 \text{ m}^3$$

$$V = 0.55 - 0.1119 \text{ m}^3$$

$$V = 0.0615 \text{ m}^3$$

$$P_R = \frac{200 \text{ bar}}{220.963 \text{ bar}} = 0.91$$

$$T_R = \frac{693 \text{ K}}{647.3 \text{ K}} = 1.07$$

$$z \approx 0.55$$

$$\% \text{ error} = \frac{|0.0776 - \text{value}|}{0.0776} \cdot 100$$

$$a) \% \text{ error} = 20.75\%$$

$$c) \% \text{ error} = 44.2\%$$

c) (10 points) (FE Exam) The roof of an electrically heated house is 7 m long, 10 m wide, and 0.25 m thick. It is made of a flat layer of concrete whose thermal conductivity is 0.92 W/m.C. During a certain winter night, the temperatures of the inner and outer surfaces of the roof are measured to be 15°C and 4°C, respectively. The average rate of heat loss through the roof that night was

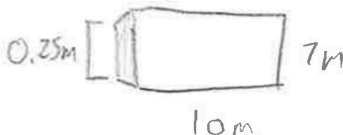
(a) 41 W

(b) 177 W

(c) 4894 W

(d) 5567 W

(e) 2834 W



$$k = 0.92 \text{ W/m}\cdot\text{C}$$

$$\Delta T = 15 - 4 = 11^\circ\text{C}$$

$$\Delta x = 0.25 \text{ m}$$

$$A = 70 \text{ m}^2$$

$$\dot{Q}_x = -kA \frac{\Delta T}{\Delta x} = -0.92 \frac{\text{W}}{\text{m}\cdot\text{C}} \cdot 70 \text{ m}^2 \cdot \left(\frac{11^\circ\text{C}}{0.25 \text{ m}} \right)$$

$$= -2833.6 \text{ W}$$

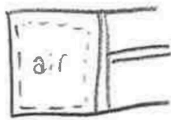
2. (35 pts)

A piston cylinder device initially contains 1.5 m^3 of air at 750 kPa and 100°C . The air is now expanded to 9 m^3 , during which the pressure-volume relationship is $PV^{1.2} = \text{constant}$. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.

- (2 pts) Draw a schematic of the system and label the system boundary. $P_1 = 7 \text{ bar}$
- (2 pts) State the engineering model $P_2 = 0.8735 \text{ bar}$
- (2 pts) Write the simplified energy balance and compressibility factor (Z).
- (4pts) Draw P-v diagram (Pressure-specific volume)
- (10 pts) Determine the work (kJ) $0.363 \text{ kmol} = n = \frac{P_1 V_1}{RT} = \frac{750 \text{ kPa} \cdot 1.5 \text{ m}^3}{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot 373 \text{ K}}$
- (15 pts) Determine the heat (kJ) (i) by using Table, (ii) by using constant specific heat values, (iii) by using $k = 1.4$

Temp. K	c_p	c_v	k
Air			
250	1.003	0.716	1.401
300	1.005	0.718	1.400
350	1.008	0.721	1.398
400	1.013	0.726	1.395
450	1.020	0.733	1.391
500	1.029	0.742	1.387
550	1.040	0.753	1.381
600	1.051	0.764	1.376
650	1.063	0.776	1.370
700	1.075	0.788	1.364
750	1.087	0.800	1.359

2.) state 1

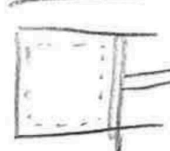


$$V_1 = 1.5 \text{ m}^3$$

$$P_1 = 750 \text{ kPa}$$

$$T_1 = 100^\circ\text{C} = 373 \text{ K}$$

state 2



$$V_2 = 9 \text{ m}^3$$

$$P_2 = 87.35 \text{ kPa}$$

$$T_2 = 260.48^\circ\text{K} = -12.51^\circ\text{C}$$

b.)

- ideal gas
- closed system
- Polytropic process
- $\Delta KE = \Delta PE = 0$

$$P_1 V_1^{1.2} = P_2 V_2^{1.2}$$

$$260.48 \text{ K} \cdot \frac{P_2 V_2}{nR} = T_2 = \frac{87.35 \text{ kPa} \cdot 9 \text{ m}^3}{0.363 \text{ kmol} \cdot 8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}}}$$

$$\frac{P_1 V_1^{1.2}}{V_2^{1.2}} = P_2 = \frac{750 (1.5)^{1.2}}{9^{1.2}} = P_2 = 87.35 \text{ kPa} \quad 0.0232$$

$$0.186 = \frac{P_1}{P_2} = \frac{7 \text{ bar}}{37.7 \text{ bar}} \quad P_2 = \frac{0.8735 \text{ bar}}{37.7 \text{ bar}}$$

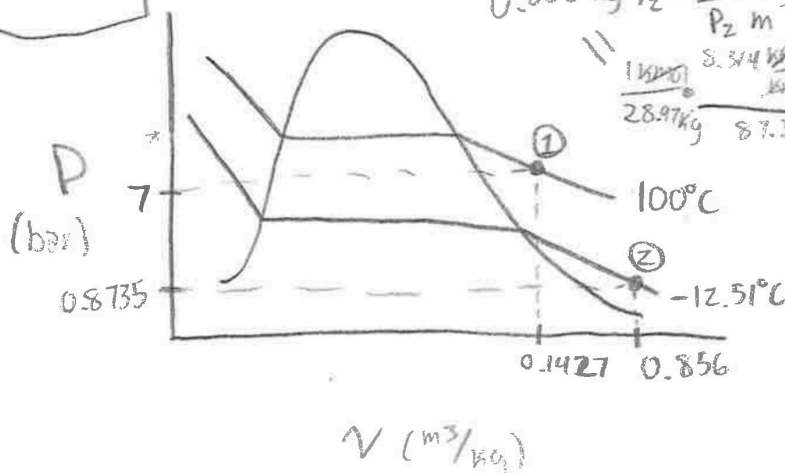
C.) $\Delta E = Q - w$
 $\Delta E = \Delta U$

$$\Delta U = Q - w$$

$$z_1 \approx 0.99$$

$$z_2 \approx 0.98$$

d.)



$$0.856 \frac{\text{m}^3}{\text{kg}} v_2 = \frac{RT_2}{P_2} = \frac{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot 260.48 \text{ K}}{87.35 \text{ kPa}} = 24.97 \frac{\text{m}^3}{\text{kmol}}$$

$$\frac{T_1}{T_2} = \frac{373 \text{ K}}{260.48 \text{ K}} = 1.43 \quad \frac{P_1}{P_2} = \frac{7 \text{ bar}}{0.8735 \text{ bar}} = 7.99$$

$$v_1 = \frac{RT_1}{P_1} = \frac{8.314 \frac{\text{kJ}}{\text{kmol} \cdot \text{K}} \cdot 373 \text{ K}}{750 \text{ kPa}} = 4.04 \frac{\text{m}^3}{\text{kmol}}$$

$$v_1 = 0.1427 \frac{\text{m}^3}{\text{kg}}$$

$$PV^{1.2} = \text{constant} = P_1 V_1^{1.2} = 750 \text{ kPa} \cdot (1.5 \text{ m}^3)^{1.2} = 1220.03$$

$$P = \frac{1220.03}{V^{1.2}}$$

$$W = \int_{V_1}^{V_2} P dV = \int_{1.5}^9 \frac{1220.03}{V^{1.2}} dV = 1694.096 \text{ kPa} \cdot \text{m}^3 = \boxed{1694.1 \text{ kJ} = W}$$

$$\Delta U = m(u_2 - u_1) = m C_v \Delta T$$

$$C_v \approx 0.721$$

$$\Delta U = 10.52 \text{ kg} (0.721 (260.48 - 373 \text{ °K}))$$

$$\Delta U = -853.46 \text{ kJ}$$

$$0.363 \text{ kmol} \times \frac{28.97 \text{ kg}}{1 \text{ kmol}} = 10.52 \text{ kg}$$

$$Q = \Delta U + W$$

ii)

$$\boxed{Q = -853.46 + 1694.1 = 840.64 \text{ kJ}}$$

iii)

$$K = 1.4$$

$$\Delta U = m C_v(T) \Delta T$$

$$= m \left(\frac{R}{K-1} \right) (-112.52 \text{ °K}), R = 0.3016$$

$$10.52 \cdot 0.754 \cdot -112.52$$

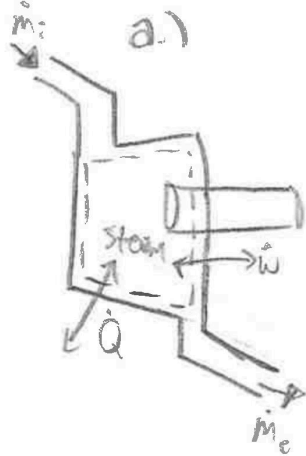
$$\Delta U = -892.51$$

$$Q = 801 \text{ kJ}$$

3. (35 points)

A steam turbine operating at steady state is providing power. Steam enters turbine at 300°C and 20 bar. The steam exits the turbine at 160°C and 3 bar. The mass flow rate of the steam at the turbine inlet is 5.5 kg/s. The rate of heat transfer from the turbine to its surroundings is 80 kJ/min. Kinetic and potential energy changes are negligible.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (2 pts) State the engineering model
- (3 pts) Write the simplified energy rate balance
- (8 pts) Determine the volumetric flowrate of the steam at the turbine exit, in m³/s.
- (10 pts) Determine the power in kW.
- (5 pts) Sketch the process from State 1 to State 2 on a P-v diagram.
- (5 pts) Explain how you would increase the power output.



$$T_i = 300^\circ\text{C} \quad T_e = 160^\circ\text{C}$$

$$P_i = 20 \text{ bar} \quad P_e = 3 \text{ bar}$$

$$\dot{m}_i = 5.5 \text{ kg/s}$$

b.)

$$\Delta KE = \Delta PE = 0$$

• Open system

• Steady state

$$\dot{m}_i = \dot{m}_e$$

• One inlet, one outlet

$$\Delta KE = \Delta PE = 0$$

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right)$$

$$\dot{Q} = -80 \frac{\text{kJ}}{\text{min}}$$

d.) Table A-4

$$0 = \dot{Q} - \dot{W} + \dot{m}_i h_i - \dot{m}_e h_e$$

$$\dot{m}_i (h_e - h_i) = \dot{Q} - \dot{W}$$

$$\left(\frac{0.907 - 0.844}{320 - 280} \right) 20 + 0.844 = 0.651 \frac{\text{m}^3}{\text{kg}} @ 160^\circ\text{C} \text{ \& } 3 \text{ bar}$$

$$\dot{m}_e v = \dot{V} = 5.5 \text{ kg/s} \cdot 0.651 \frac{\text{m}^3}{\text{kg}} = \boxed{3.581 \frac{\text{m}^3}{\text{s}}} \text{ d.)}$$

e.)

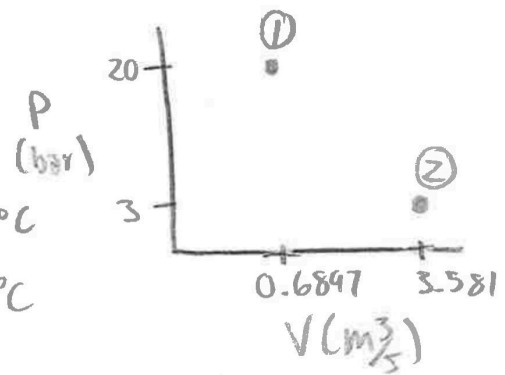
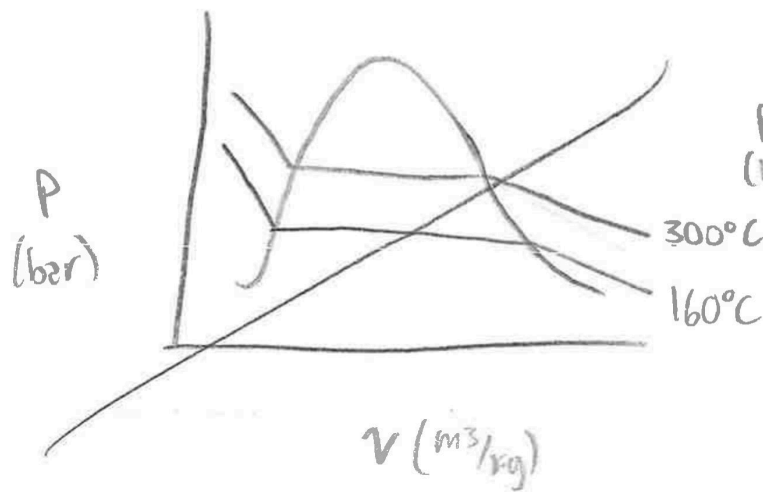
$$\dot{W} = \dot{Q} - \dot{m}_i (h_e - h_i)$$

$$h_i = \left(\frac{3069.5 - 2976.4}{320 - 280} \right) \cdot 20 + 2976.4 = 3022.95 \frac{\text{kJ}}{\text{kg}}$$

$$-80 \frac{\text{kJ}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = -1.33 \frac{\text{kJ}}{\text{s}}$$

$$h_e = 2782.3 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{W} = -1.33 \frac{\text{kJ}}{\text{s}} - 5.5 \frac{\text{kg}}{\text{s}} (2782.3 \frac{\text{kJ}}{\text{kg}} - 3022.95 \frac{\text{kJ}}{\text{kg}}) = \dot{W} = \boxed{P = 1322.24 \text{ kW}} \text{ e.)}$$



don't know the path of process

$$v_i = \left(\frac{0.1308 - 0.1200}{320 - 280} \right) 20 + 0.1200 = 0.1254 \text{ m}^3/\text{kg}, 300^\circ\text{C}; 20 \text{ bar}$$

$$v_e = 0.651 \text{ m}^3/\text{kg}, 160^\circ\text{C}, 3 \text{ bar}, v_e \cdot 5.5 \frac{\text{kg}}{\text{s}} = 3.581 \text{ m}^3/\text{s} = \frac{v_e}{+}$$

$$\frac{v_i}{-} = v_i \cdot 5.5 \frac{\text{kg}}{\text{s}} = 0.6847 \text{ m}^3/\text{s}$$

• 9.) To increase the power output, I would increase the inlet/outlet mass flow rate, and keep the system in a cold environment to increase the change in enthalpy.